

Unit-I

Class-B.Sc(ECS)-II

Subject-Operating System

1. Operating System Basics

1. What is an Operating System?

- A) Application software
- B) System software that manages computer resources
- C) Programming language
- D) Hardware device

Answer: B

2. Which of the following is an example of an Operating System?

- A) MS Word
- B) Google Chrome
- C) Linux
- D) Photoshop

Answer: C

3. The primary purpose of an Operating System is to:

- A) Create documents
- B) Manage hardware and software resources
- C) Browse the internet
- D) Write programs only

Answer: B

4. Which of the following is NOT an operating system?

- A) Windows
- B) Linux
- C) macOS
- D) Oracle

Answer: D

5. The OS acts as an interface between:

- A) User and hardware
- B) Hardware and electricity
- C) Compiler and programmer
- D) Keyboard and mouse

Answer: A

6. In a Batch Operating System, jobs are:

- A) Executed interactively

- B) Grouped and executed in batches
- C) Always executed simultaneously
- D) Executed only by the user

Answer: B

7. Which OS allows several programs to reside in memory at the same time?

- A) Batch OS
- B) Multiprogramming OS
- C) Single-user OS
- D) Embedded OS

Answer: B

8. The main objective of multiprogramming is to:

- A) Reduce CPU utilization
- B) Increase CPU utilization
- C) Increase keyboard speed
- D) Reduce memory size

Answer: B

9. A Time-Sharing Operating System provides:

- A) CPU time to users/processes in small time slices
- B) Only one program at a time
- C) No user interaction
- D) Only batch processing

Answer: A

10. Which operating system is designed to respond within a specified time limit?

- A) Batch OS
- B) Real-Time OS
- C) Distributed OS
- D) Time-Sharing OS

Answer: B

11. Which is an important characteristic of a Real-Time Operating System?

- A) Guaranteed response within timing constraints
- B) Very slow response
- C) No process management
- D) No memory management

Answer: A

12. A Distributed Operating System manages:

- A) A single CPU only
- B) Multiple connected computers as a coordinated system

- C) Only input devices
- D) Only printers

Answer: B

13. A Parallel Operating System is mainly designed to support:

- A) Multiple processors working together
- B) A single-user environment
- C) Only batch jobs
- D) Only mobile devices

Answer: A

14. Which OS type is commonly used where immediate response is critical?

- A) Batch OS
- B) Real-Time OS
- C) Distributed OS
- D) Offline OS

Answer: B

15. Which OS type allows multiple users to interact with a computer system simultaneously?

- A) Time-Sharing OS
- B) Batch OS
- C) Single-tasking OS
- D) Embedded OS

Answer: A

16. Which of the following is an Operating System service?

- A) Program execution
- B) Process management
- C) File management
- D) All of the above

Answer: D

17. File management is responsible for:

- A) Managing files and directories
- B) Designing hardware
- C) Creating processors
- D) Increasing CPU voltage

Answer: A

18. Which OS service manages the allocation of memory to programs?

- A) Memory management
- B) File management

- C) Security management
- D) User interface

Answer: A

19. Protection and security services are used to:

- A) Prevent unauthorized access
- B) Increase monitor size
- C) Reduce keyboard usage
- D) Format documents

Answer: A

20. An OS provides an interface through which users can interact with the system. This is called:

- A) User interface
- B) Compiler
- C) Loader
- D) Assembler

Answer: A

21. Which component manages processes in an operating system?

- A) Process management component
- B) Keyboard
- C) Monitor
- D) Compiler

Answer: A

22. Which component manages the allocation and deallocation of memory?

- A) File management
- B) Memory management
- C) I/O management
- D) Security management

Answer: B

23. I/O management is responsible for:

- A) Managing input/output devices
- B) Managing only files
- C) Managing only processes
- D) Writing source code

Answer: A

24. Which component manages files and directories?

- A) Process manager
- B) Memory manager

- C) File system
- D) CPU scheduler

Answer: C

25. Secondary storage management deals with:

- A) Hard disks and other storage devices
- B) CPU registers only
- C) Keyboard input only
- D) Monitor resolution

Answer: A

26. A system call provides an interface between:

- A) User program and operating system
- B) Hardware and electricity
- C) Monitor and keyboard
- D) Compiler and printer

Answer: A

27. Which of the following is an example of a system call?

- A) open()
- B) printf() only
- C) HTML
- D) CSS

Answer: A

28. System calls are primarily used by:

- A) Application programs to request OS services
- B) Monitors to display images
- C) Keyboards to send power
- D) Printers to store files

Answer: A

29. Which system call category is used to create or terminate processes?

- A) Process control
- B) File management
- C) Communication
- D) Information maintenance

Answer: A

30. Which of the following is related to file management system calls?

- A) open()
- B) fork()
- C) exec()

D) wait()

Answer: A

31. A process is:

- A) A program in execution
- B) A program stored on disk only
- C) A hardware component
- D) A file extension

Answer: A

32. Which of the following is an essential part of a process?

- A) Program counter
- B) CPU registers
- C) Stack
- D) All of the above

Answer: D

33. The program counter of a process contains:

- A) Address of the next instruction to execute
- B) Total memory size
- C) Process name only
- D) File size

Answer: A

34. Process management mainly deals with:

- A) Creation, scheduling, and termination of processes
- B) Designing websites
- C) Formatting documents
- D) Manufacturing CPUs

Answer: A

35. Which of the following is a common process state?

- A) New
- B) Ready
- C) Running
- D) All of the above

Answer: D

36. A process that is currently being executed by the CPU is in the:

- A) Ready state
- B) Waiting state
- C) Running state

D) New state

Answer: C

37. A process waiting for CPU allocation is in the:

A) Running state

B) Ready state

C) Terminated state

D) New state

Answer: B

38. A process waiting for an I/O operation to complete is generally in the:

A) Waiting/Blocked state

B) Running state

C) Ready state

D) New state

Answer: A

39. After completing execution, a process enters the:

A) Ready state

B) Waiting state

C) Terminated state

D) New state

Answer: C

40. PCB stands for:

A) Program Control Block

B) Process Control Block

C) Process Communication Buffer

D) Program Communication Block

Answer: B

41. The PCB contains:

A) Process state

B) Program counter

C) CPU registers

D) All of the above

Answer: D

42. Which information is NOT normally stored in a PCB?

A) Process state

B) CPU registers

C) Program counter

D) User's favorite color

Answer: D

43. The PCB is maintained by:

A) Operating System

B) User

C) Compiler only

D) Printer

Answer: A

44. Context switching means:

A) Switching from one process to another

B) Changing the operating system

C) Changing the computer's hardware

D) Restarting the computer

Answer: A

45. During a context switch, the OS saves:

A) Current process state

B) Monitor settings only

C) Keyboard layout only

D) Printer settings only

Answer: A

46. Context switching is necessary when:

A) CPU changes from one process to another

B) A file is renamed

C) A monitor is turned off

D) A keyboard is disconnected

Answer: A

47. Context switching generally causes:

A) Some CPU overhead

B) No overhead

C) Permanent CPU failure

D) More disk space

Answer: A

48. Which operation creates a new process?

A) Process creation

B) Process deletion

C) Process blocking

D) Process termination

Answer: A

49. Which operation removes a process from the system?

A) Process creation

B) Process termination

C) Process scheduling

D) Context switching

Answer: B

50. The fork() system call in Unix/Linux is commonly used to:

A) Create a new process

B) Delete a file

C) Open a printer

D) Allocate a hard disk

Answer: A

51. A parent process can create:

A) Child processes

B) Only files

C) Only threads

D) Hardware devices

Answer: A

52. A process may be blocked when it:

A) Waits for an event or I/O operation

B) Is executing normally

C) Has just been created

D) Has terminated

Answer: A

53. Which operation changes the state of a process from running to waiting?

A) Waiting for I/O

B) Process creation

C) Termination

D) Compilation

Answer: A

Unit-II

1. What is process scheduling?

A) Creating files

B) Selecting a process for CPU execution

C) Managing memory only

D) Managing I/O devices only

Answer: B

2. The scheduler that selects a process from the ready queue is called:

- A) Long-term scheduler
- B) Medium-term scheduler
- C) Short-term scheduler
- D) Job scheduler

Answer: C

3. The long-term scheduler is also known as:

- A) CPU scheduler
- B) Job scheduler
- C) Memory scheduler
- D) I/O scheduler

Answer: B

4. Which scheduler controls the degree of multiprogramming?

- A) Short-term scheduler
- B) Long-term scheduler
- C) I/O scheduler
- D) CPU scheduler

Answer: B

5. The medium-term scheduler is mainly associated with:

- A) Compilation
- B) Swapping
- C) File creation
- D) CPU execution

Answer: B

6. Which scheduler executes most frequently?

- A) Long-term scheduler
- B) Medium-term scheduler
- C) Short-term scheduler
- D) Job scheduler

Answer: C

7. Which scheduling criterion measures the amount of time the CPU is busy?

- A) Turnaround time
- B) Waiting time
- C) CPU utilization

D) Response time

Answer: C

8. Throughput refers to:

A) Time taken by one process

B) Number of processes completed per unit of time

C) Time spent waiting

D) Time spent in I/O

Answer: B

9. Turnaround time is the time between:

A) Process creation and first response

B) Process submission and process completion

C) Two CPU bursts

D) Two I/O operations

Answer: B

10. Waiting time is the total time a process spends:

A) Executing on CPU

B) In the ready queue

C) Performing I/O

D) Terminated

Answer: B

11. Response time is the time between:

A) Submission of a request and the first response

B) Process creation and termination

C) Two CPU bursts

D) Start and completion of I/O

Answer: A

12. Which of the following is generally desirable in CPU scheduling?

A) High waiting time

B) High turnaround time

C) High CPU utilization

D) Low throughput

Answer: C

13. In preemptive scheduling:

A) CPU cannot be taken from a process

B) CPU can be taken from a running process

C) Processes never wait

D) Only one process exists

Answer: B

14. Which is a preemptive scheduling algorithm?

A) FCFS

B) Non-preemptive SJF

C) Round Robin

D) Non-preemptive Priority

Answer: C

15. In non-preemptive scheduling, a process normally retains the CPU until:

A) Another process arrives

B) It completes or blocks

C) Its priority changes

D) Memory becomes full

Answer: B

16. Which scheduling algorithm is normally non-preemptive?

A) Round Robin

B) FCFS

C) SRTF

D) Preemptive Priority

Answer: B

17. FCFS stands for:

A) First Come First Served

B) First CPU First Served

C) First Control First Service

D) Fast Come Fast Served

Answer: A

18. FCFS schedules processes according to their:

A) Priority

B) Burst time

C) Arrival order

D) Memory size

Answer: C

19. FCFS is generally a:

A) Preemptive algorithm

B) Non-preemptive algorithm

C) Real-time algorithm

D) Parallel algorithm

Answer: B

20. A major disadvantage of FCFS is:

A) Starvation always occurs

B) Convoy effect

C) No process can execute

D) It requires multiple CPUs

Answer: B

21. SJF stands for:

A) Shortest Job First

B) Smallest Job Fast

C) Short Job Function

D) System Job First

Answer: A

22. SJF selects the process with the:

A) Highest priority

B) Longest burst time

C) Shortest burst time

D) Earliest completion time

Answer: C

23. SJF can be implemented as:

A) Only preemptive

B) Only non-preemptive

C) Both preemptive and non-preemptive

D) Neither

Answer: C

24. The preemptive version of SJF is called:

A) FCFS

B) Round Robin

C) SRTF

D) Multilevel Queue

Answer: C

25. SRTF stands for:

A) Shortest Remaining Time First

B) Shortest Running Task First

C) System Response Time First

D) Short Response Task First

Answer: A

26. A disadvantage of SJF is:

A) It cannot execute processes

B) Long processes may experience starvation

C) It does not use CPU

D) It requires no scheduling

Answer: B

27. Round Robin scheduling is mainly used in:

A) Batch systems

B) Time-sharing systems

C) Offline systems

D) Single-task systems

Answer: B

28. Round Robin uses a fixed time interval called:

A) Burst time

B) Waiting time

C) Time quantum

D) Turnaround time

Answer: C

29. Round Robin is a:

A) Non-preemptive algorithm

B) Preemptive algorithm

C) Static algorithm only

D) Memory allocation algorithm

Answer: B

30. If a process does not finish within its time quantum, it is:

A) Terminated

B) Removed permanently

C) Moved to the ready queue

D) Deleted

Answer: C

31. If the time quantum is very large, Round Robin behaves similarly to:

A) SJF

B) FCFS

C) Priority Scheduling

D) SRTF

Answer: B

32. In Priority Scheduling, the CPU is allocated based on:

A) Arrival time only

B) Priority of the process

C) Memory size

D) Process name

Answer: B

33. Priority Scheduling can be:

A) Only preemptive

B) Only non-preemptive

C) Both preemptive and non-preemptive

D) Neither

Answer: C

34. A major problem with Priority Scheduling is:

A) Deadlock always occurs

B) Starvation

C) No process can execute

D) CPU cannot be used

Answer: B

35. The technique used to prevent starvation in Priority Scheduling is:

A) Paging

B) Aging

C) Swapping

D) Fragmentation

Answer: B

36. Aging means:

A) Decreasing the priority of waiting processes

B) Gradually increasing the priority of waiting processes

C) Deleting old processes

D) Increasing CPU speed

Answer: B

37. In Multilevel Queue Scheduling, the ready queue is:

A) Divided into multiple queues

B) Always empty

C) Converted into a stack

D) Deleted

Answer: A

38. Different queues in Multilevel Queue Scheduling may use:

A) Different scheduling algorithms

B) Only FCFS

C) Only SJF

D) No scheduling algorithm

Answer: A

39. In traditional Multilevel Queue Scheduling, a process generally:

A) Moves randomly between queues

B) Remains permanently assigned to a queue

C) Is immediately terminated

D) Cannot execute

Answer: B

40. The main difference between Multilevel Queue and Multilevel Feedback Queue is that:

A) MLFQ allows processes to move between queues

B) MLFQ has no CPU

C) Multilevel Queue has no processes

D) Both are exactly the same

Answer: A

41. In Multilevel Feedback Queue, a process may move to a lower-priority queue when it:

A) Uses excessive CPU time

B) Terminates

C) Creates a file

D) Performs no operation

Answer: A

42. Multilevel Feedback Queue is designed to:

A) Provide flexible process scheduling

B) Eliminate CPU usage

C) Manage files only

D) Replace memory

Answer: A

43. Process synchronization is required mainly to:

A) Increase disk size

B) Coordinate processes accessing shared resources

C) Create hardware

D) Compile programs

Answer: B

44. A Race Condition occurs when:

A) Only one process executes

B) The result depends on the order of concurrent access to shared data

C) CPU is turned off

D) Memory is empty

Answer: B

45. The part of a program where shared data is accessed is called:

A) Entry section

B) Critical section

C) Exit section

D) Waiting section

Answer: B

46. Which is NOT a requirement of a correct critical-section solution?

A) Mutual exclusion

B) Progress

C) Bounded waiting

D) Infinite waiting

Answer: D

47. Mutual exclusion means:

A) Multiple processes must enter the critical section together

B) Only one process can execute in the critical section at a time

C) No process can enter the critical section

D) All processes must terminate

Answer: B

48. A semaphore is used for:

A) Process synchronization

B) File compression

C) Virus detection

D) Disk formatting

Answer: A

49. The two common semaphore operations are:

A) Read and Write

B) Open and Close

C) Wait and Signal

D) Start and Stop

Answer: C

50. A binary semaphore generally has values:

A) 0 and 1

B) 1 and 2

C) 0 to 100

D) Only 10

Answer: A

51. The Producer-Consumer problem involves a shared:

A) CPU

B) Buffer

C) Compiler

D) Keyboard

Answer: B

52. In the Producer-Consumer problem, the producer:

A) Consumes data

B) Produces data and places it in the buffer

C) Deletes the buffer

D) Terminates the consumer

Answer: B

53. In the Producer-Consumer problem, the consumer should not remove data when the buffer is:

A) Full

B) Empty

C) Half full

D) Large

Answer: B

54. The Reader-Writer problem deals with access to:

A) Shared data

B) CPU registers only

C) Keyboard only

D) Printer only

Answer: A

55. In the Reader-Writer problem:

A) Multiple readers can generally read simultaneously

B) Only one reader can ever read

C) Readers cannot access data

D) Writers never access data

Answer: A

56. A writer generally requires:

- A) Shared access with other writers
- B) Exclusive access to the shared data
- C) No access to data
- D) Only read access

Answer: B

57. The Dining Philosophers problem is mainly related to:

- A) Resource sharing and synchronization
- B) File management
- C) Memory allocation only
- D) CPU architecture

Answer: A

58. In the Dining Philosophers problem, each philosopher needs how many forks to eat?

- A) One
- B) Two
- C) Three
- D) Four

Answer: B

59. If every philosopher holds one fork and waits for another, the system may experience:

- A) Starvation only
- B) Deadlock
- C) Compilation
- D) Paging

Answer: B

60. The Dining Philosophers problem demonstrates problems related to:

- A) Synchronization and resource allocation
- B) File compression
- C) Computer graphics
- D) Network addressing

Answer: A